



Indian Institute of Technology Madras

EDUCATION

PROGRAM	INSTITUTION	% / CGPA	COMPLETION
B Tech, Civil Engineering	Indian Institute of Technology Madras		2021
Class XII	Narayana Junior college		2017
Class X	Narayana E-techno school		2015

SCHOLASTIC ACHIEVEMENTS

- Ranked amongst **Top 1.8%** in JEE-Advanced (**out of 1,55,000+ candidates**) and **Top 0.2%** in JEE-Mains (out of **11,85,000+ candidates**) 2017
- Secured State Rank **401** in Telangana Eamcet and **617** in Andhra Pradesh Eamcet 2017 (out of **1,50,000+ candidates**)
- Obtained **State 1st rank** in Telangana 11th board exams (out of **4,00,000+ candidates**)

PROFESSIONAL EXPERIENCE

Software Intern, Advanced Data Processing Research Institute (ADRIN) Dec 2019 - Feb 2020

- Built an **Android app** using **Java** to capture the video of surroundings and used **TCP** sockets for data transfer between app and server
- Generated depth maps from the video using **monodepth2**. Created **Point clouds** from video and depth maps using **Open3d**

PROJECTS

Crowd Analysis — Surveillance Automation May 2018 - Jan 2019

- Implemented deep learning architectures like **ResnetCrowd**, **CSRNET** using **TensorFlow** and **Keras** for predicting real-time density maps
- Adapted the real-world anomaly detection in surveillance videos and the social force model for **abnormal behavior detection**
- Invited by the **Indian Army** to present the project in Army Tech Seminar at Manekshaw Centre, New Delhi 2019

RL Algorithms — Course project Jan 2019 - May 2020

- Created **OpenAI gym** environments **Puddle World** and **Four Rooms**. Experimented with one step SMDP Q-learning on Four Rooms
- Implemented various algorithms like SARSA, SARSA(λ), Monte-Carlo policy gradient and linear function approximation on Puddle World
- Conducted experiments on **10-arm bandit** testbed to analyze the performance of ϵ -greedy, softmax, UCB1

Self driving in MADRaS — Guide: Prof. B. Ravindran May 2020 - July 2020

- Implemented **PPO** and conducted experiments on MADRaS (built on top of TORCS), to train an RL agent to drive on different road tracks
- Trained agent on e-track-2. The test lap completion rates were **100%** on tracks alpine-1 and forza, **93.5%** on alpine-2 and **68.4%** on corkscrew
- Designed various **reward functions** to decrease the training time of the agent. Created methods to **visualize** the training and testing process

DeepDrive — Autonomous driving in CARLA Simulator June 2019 - Nov 2019

- Collaborated with a team of 5 to develop autonomous car in **CARLA** and demonstrated this at CFI Open house with a footfall of 4000 people
- Worked on predicting **distances** of objects, captured by front camera of the car using **YOLOv3**, **Semantic Segmentation** and **Depth maps**

Buy and Sell — Android app May 2018

- Developed an android app which allows vendors to display their products, change their quantity, prices and customers to buy them
- Created **product display**, **product management**, purchase summary, vendors list and signup pages. Used **SQLite database** for data storage

POSITIONS OF RESPONSIBILITY

Web Operations coordinator — Saarang Nov 2018 - Jan 2019

- Collaborated with a team of 10 to make an **IOS app** using **Ionic**. Added Payment gateway (using **RazorPay**) and Profile page to the IOS app

RELEVANT COURSEWORK

- Data Structures and Algorithms*, Object Oriented Programming*, Foundations of Computer System Design*, Reinforcement Learning, Probability and Statistics, Graph Theory, Machine Learning, Transportation Network Analysis

**ongoing courses*

SKILLS

- **Languages** - Python, C/C++, Java, Javascript
- **Libraries** - PyTorch, Numpy, OpenCV, TensorFlow, Pandas
- **Web** - HTML, CSS, Ionic
- **Software** - Android Studio, Git, Linux, $\LaTeX$, TORCS

EXTRA-CURRICULAR ACTIVITIES

- Participated in building Autonomous Cleaning Bots by CFI, IIT Madras. Cracked **Asian Book of Records** and **Indian Book of Records** 2017
- Won **Gold** and **Silver** medals in Freshie Schroeter **Cricket** and **Hockey** in 2017 respectively
- Participated in Yet Another **Hackathon** 2K18, made an online **voting** system which uses 2 level authentication using Aadhar card and OTP
- Represented Brahmaputra Hostel in **Inter Hostel Hockey** Schroeter 2018